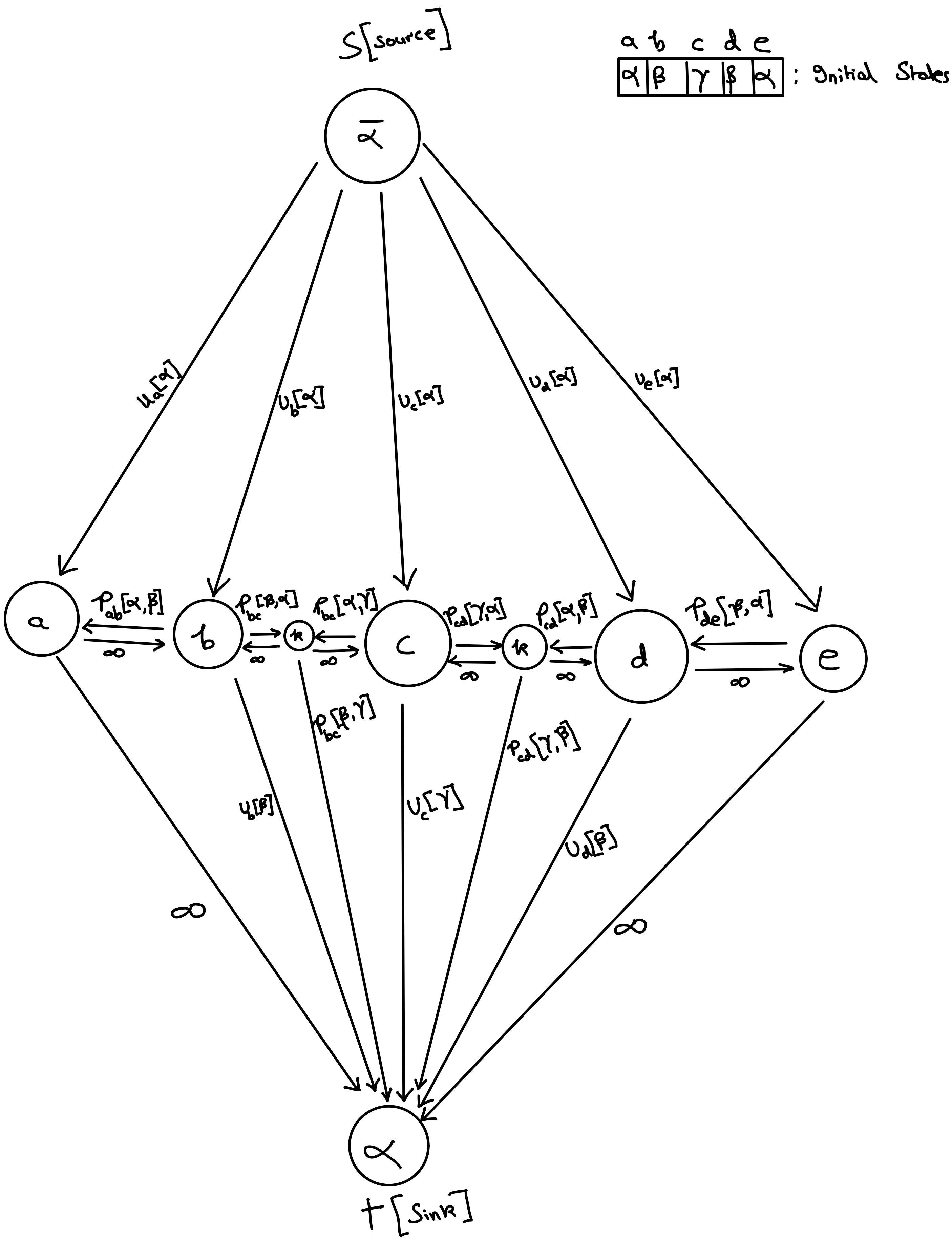
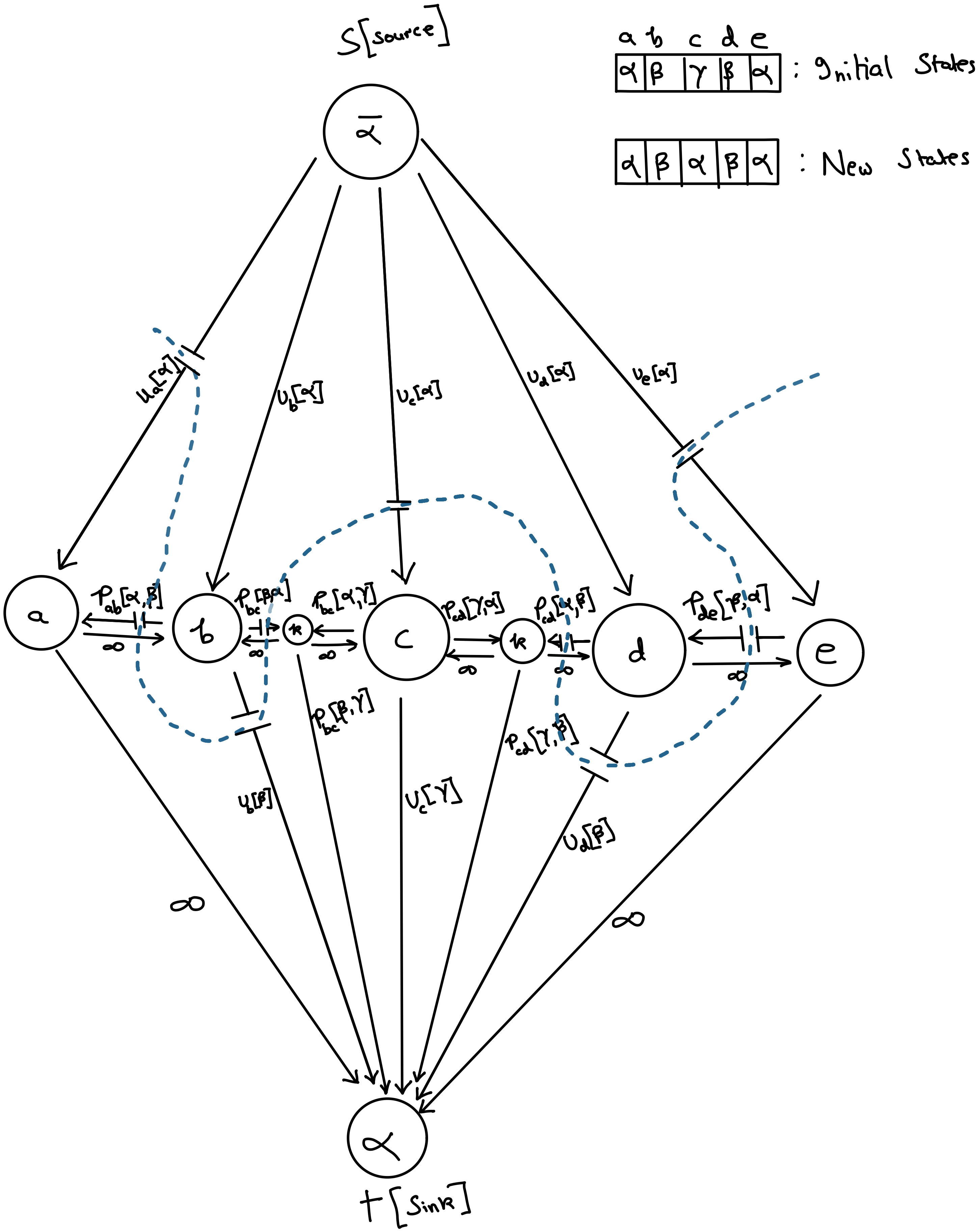
Q1:


S [source]
a b c d e
α β γ β α
: Initial States
ᾱ
$U_a[\alpha]$
$U_b[\alpha]$
$U_c[\alpha]$
$U_d[\alpha]$
$U_e[\alpha]$
a
b
c
d
e
k
k
$P_{ab}[\alpha,\beta]$
$P_{bc}[\beta,\alpha]$
$P_{bc}[\alpha,\gamma]$
$P_{cd}[\gamma,\alpha]$
$P_{cd}[\alpha,\beta]$
$P_{de}[\beta,\alpha]$
$P_{bc}[\beta,\gamma]$
$P_{cd}[\gamma,\beta]$
$U_b[\beta]$
$U_c[\gamma]$
$U_d[\beta]$
∞
α
t [Sink]



S[source]
a b c d e
α β γ β α : Initial States
α β α β α : New States
ᾱ
U_a[α]
U_b[α]
U_c[α]
U_d[α]
U_e[α]
P_ab[α,β]
P_bc[β,α]
P_bc[α,γ]
P_cd[γ,α]
P_cd[α,β]
P_de[β,α]
P_bc[β,γ]
P_cd[γ,β]
U_b[β]
U_c[γ]
U_d[β]
∞
a
b
c
d
e
k
α
t[Sink]

The mathematical expression for the total cost of this configuration $\{\alpha, \beta, \alpha, \beta, \alpha\} = L$

$$E[L] = U_a[\alpha] + U_b[\beta] + U_c[\alpha] + U_d[\beta] + U_e[\alpha] + P_{ab}[\alpha, \beta] + P_{bc}[\beta, \alpha] + P_{cd}[\alpha, \beta] + P_{de}[\beta, \alpha]$$

Report Q2:

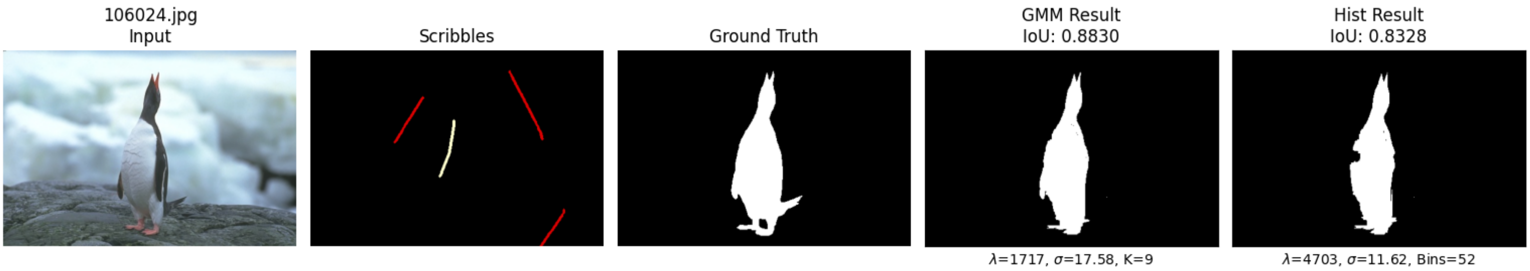
The objective was to determine which method more accurately isolates the foreground from the background across a dataset of images using the parametric Gaussian mixture model and Color Histogram. The color histogram method achieved an average IoU of 0.7529, while the GMM method achieved an average IoU of 0.7498 on the given dataset. Color histogram method provided a slightly more accurate segmentation than the GMM.

The IoU score across both methods were influenced by using specific parameter values to match individual image complexity. For GMM, the number of component [K] was critical, the model maximized accuracy on uniform objects by simplifying the distribution to a fewer components whereas the complex color variance required a higher value of K. The color histogram's accuracy relied on the quantization level [bins], using a low bin count aggregated tonal regions to reduce noise, while a high bin count preserved necessary details.

Applying high lambda values enforce strict, smooth boundaries of distinct foreground, while lower values to allow for the intricate edges.

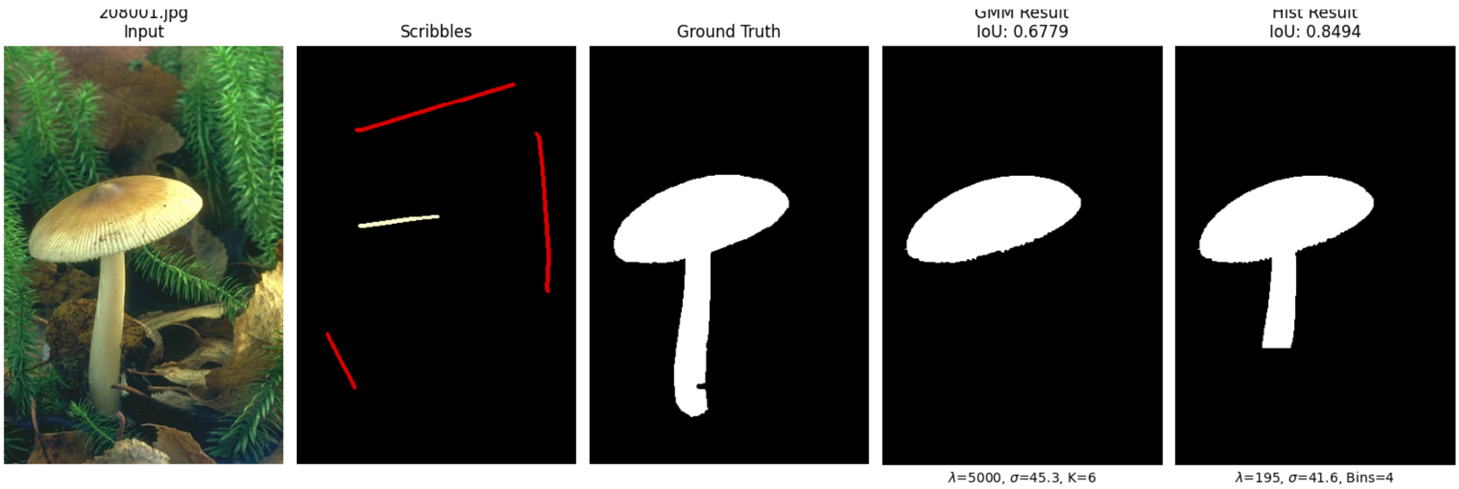
The sigma parameter controls the contrast sensitivity of the pairwise smoothness term, low sigma values were required for images with subtle boundaries to heighten edge sensitivity, whereas high sigma values allowed the model to ignore texture noise in high contrast images.

Core Graph Cut Algorithm

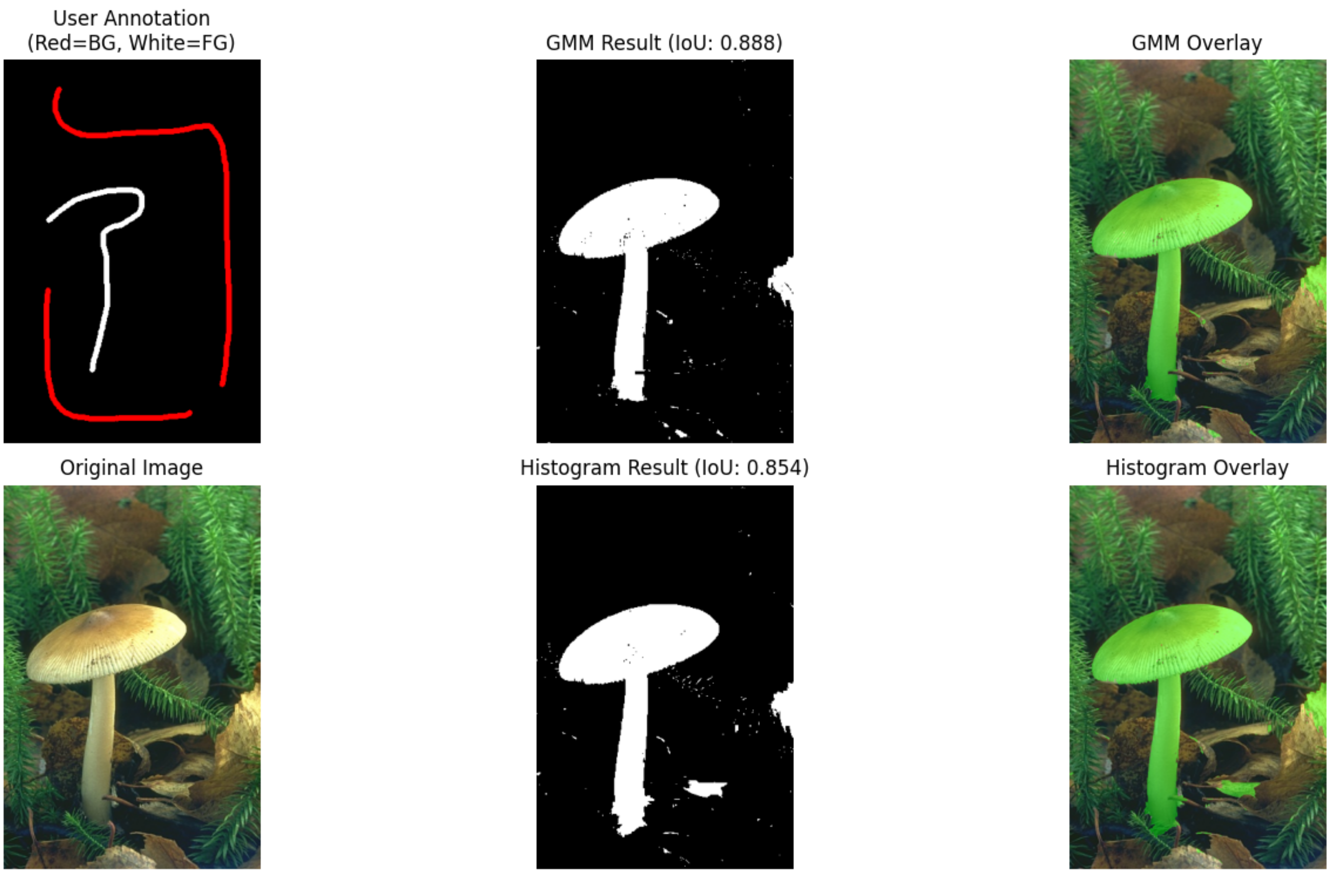


Interactive Tool

Core Graph Cut Algorithm



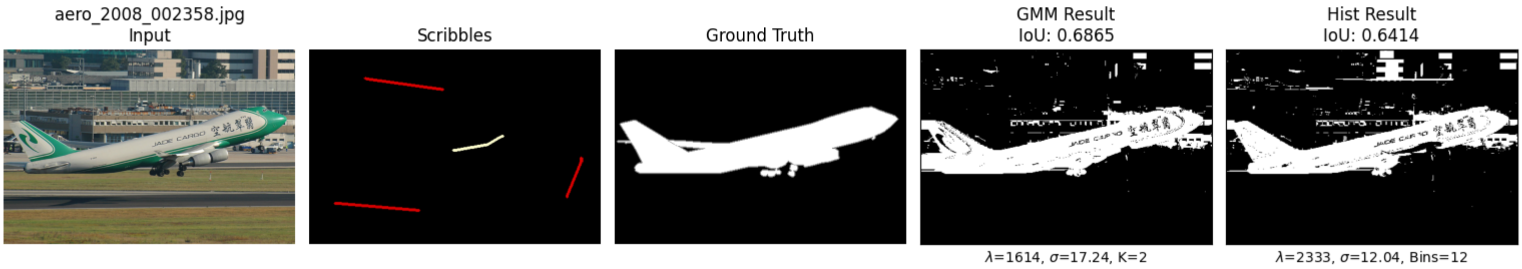
Interactive Tool



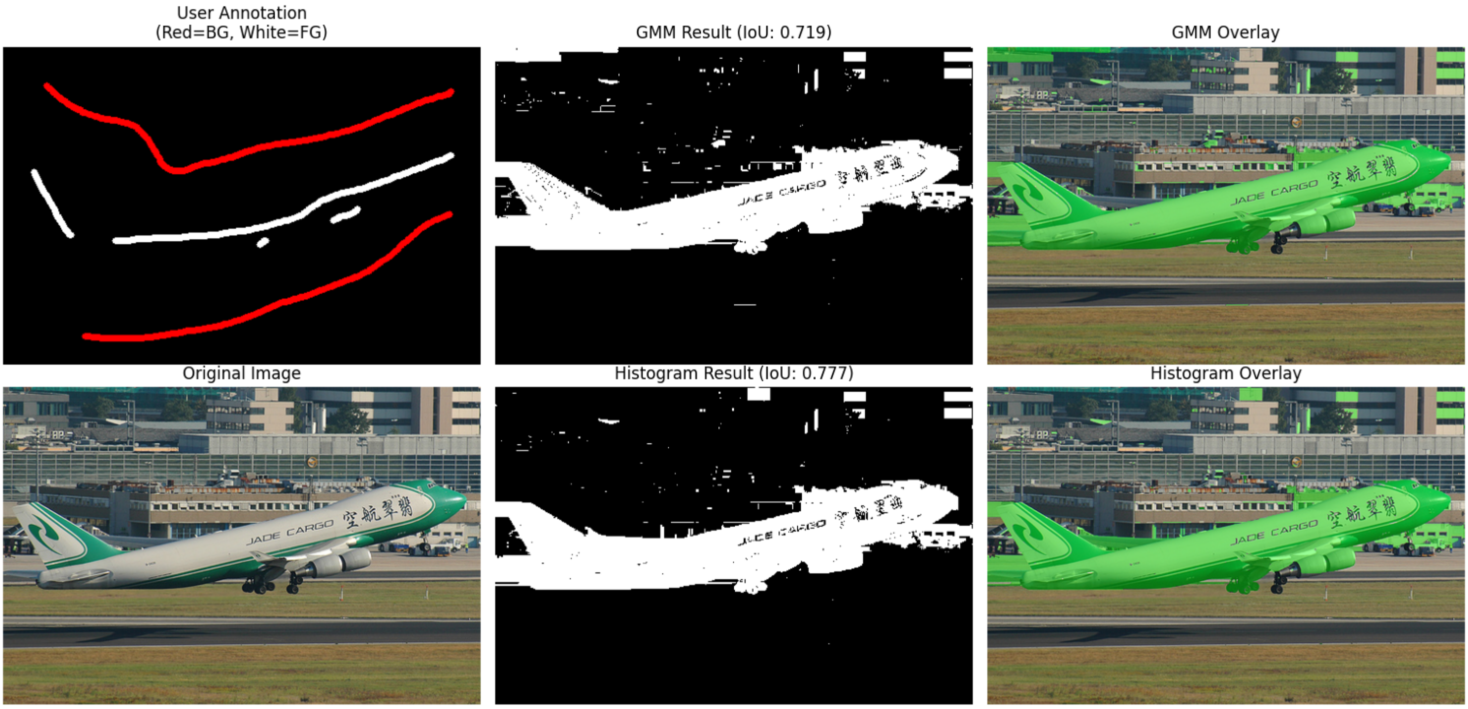
Core Graph Cut Algorithm

Interactive Tool

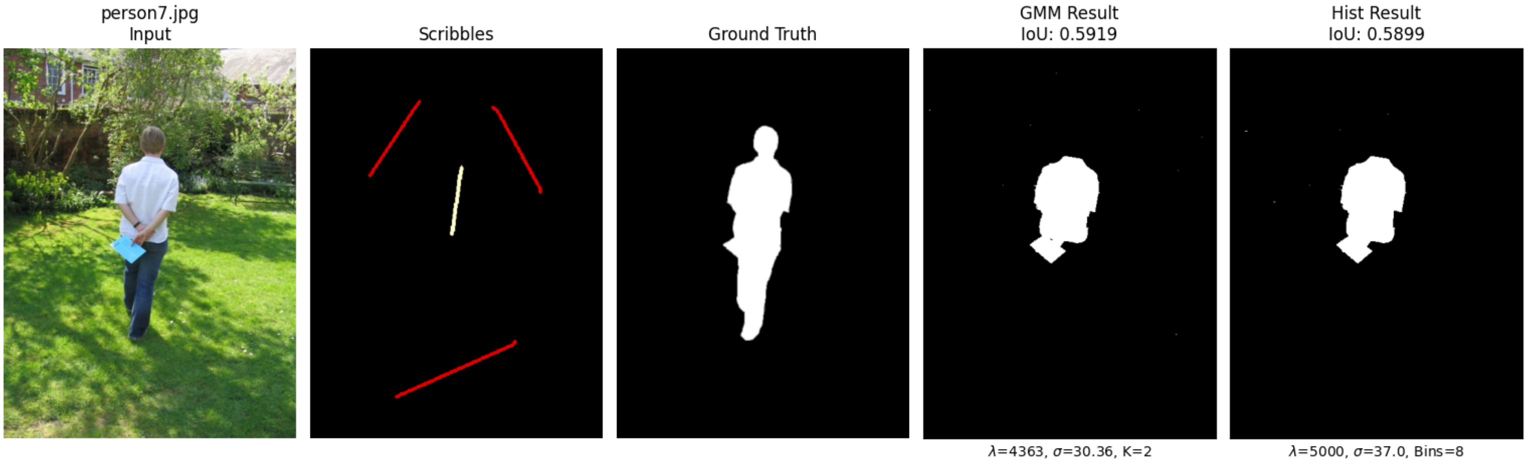
Core Graph Cut Algorithm



Interactive Tool

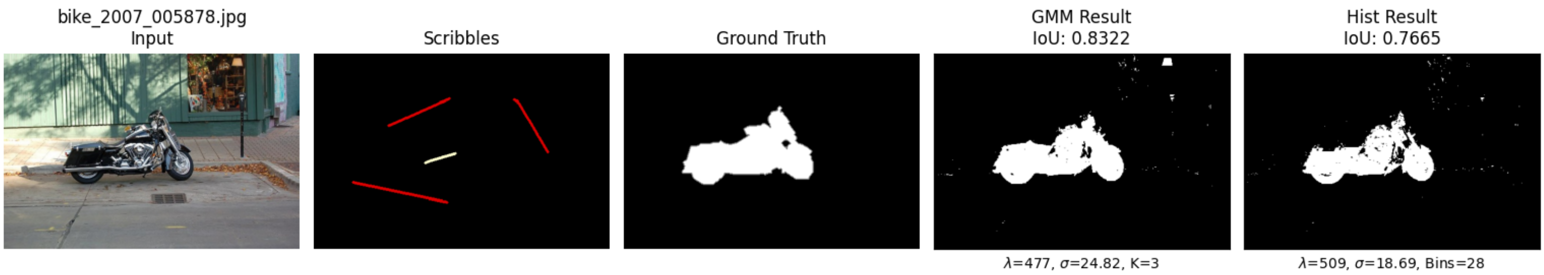


Core Graph Cut Algorithm



Interactive Tool

Core Graph Cut Algorithm



Interactive Tool